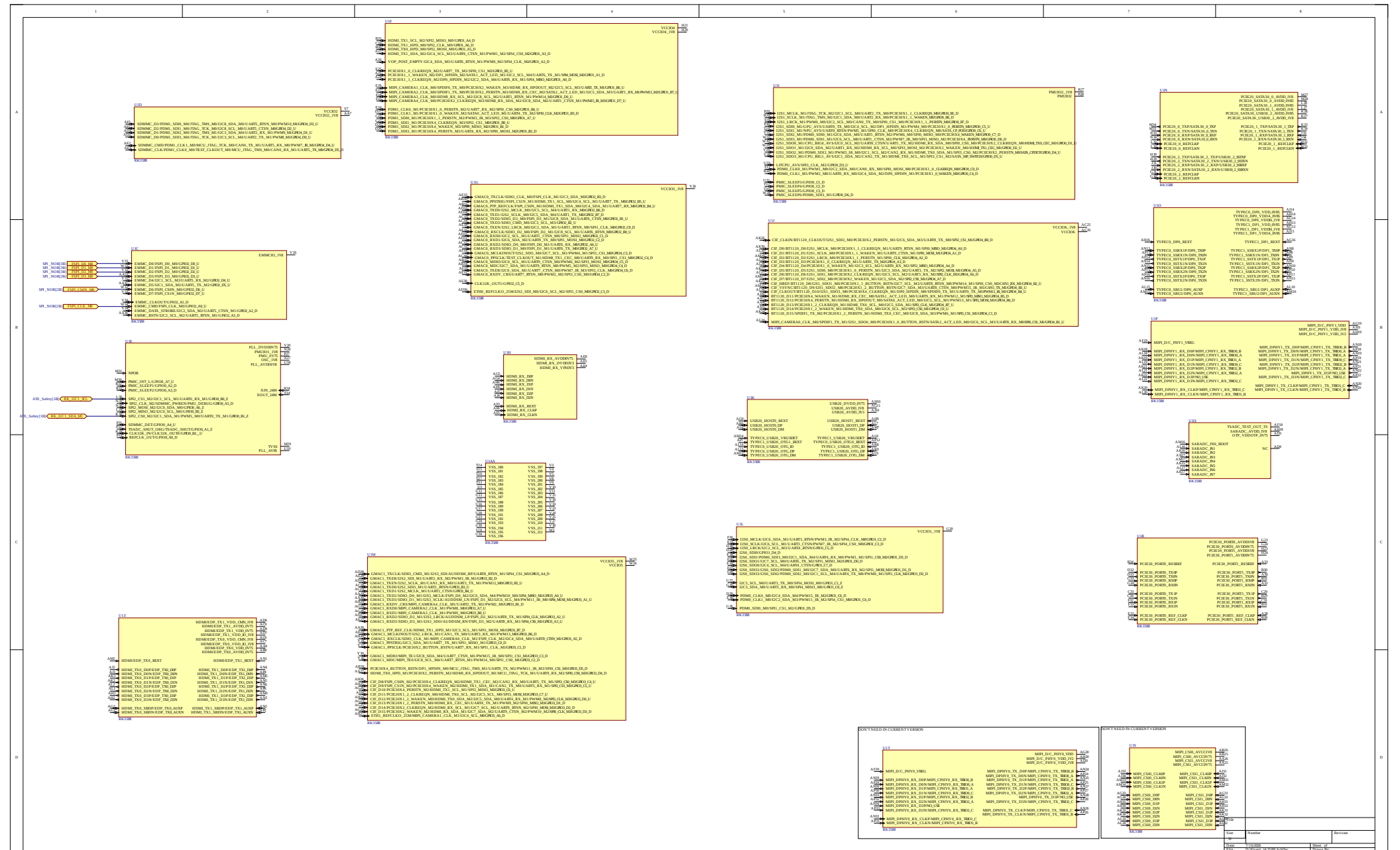
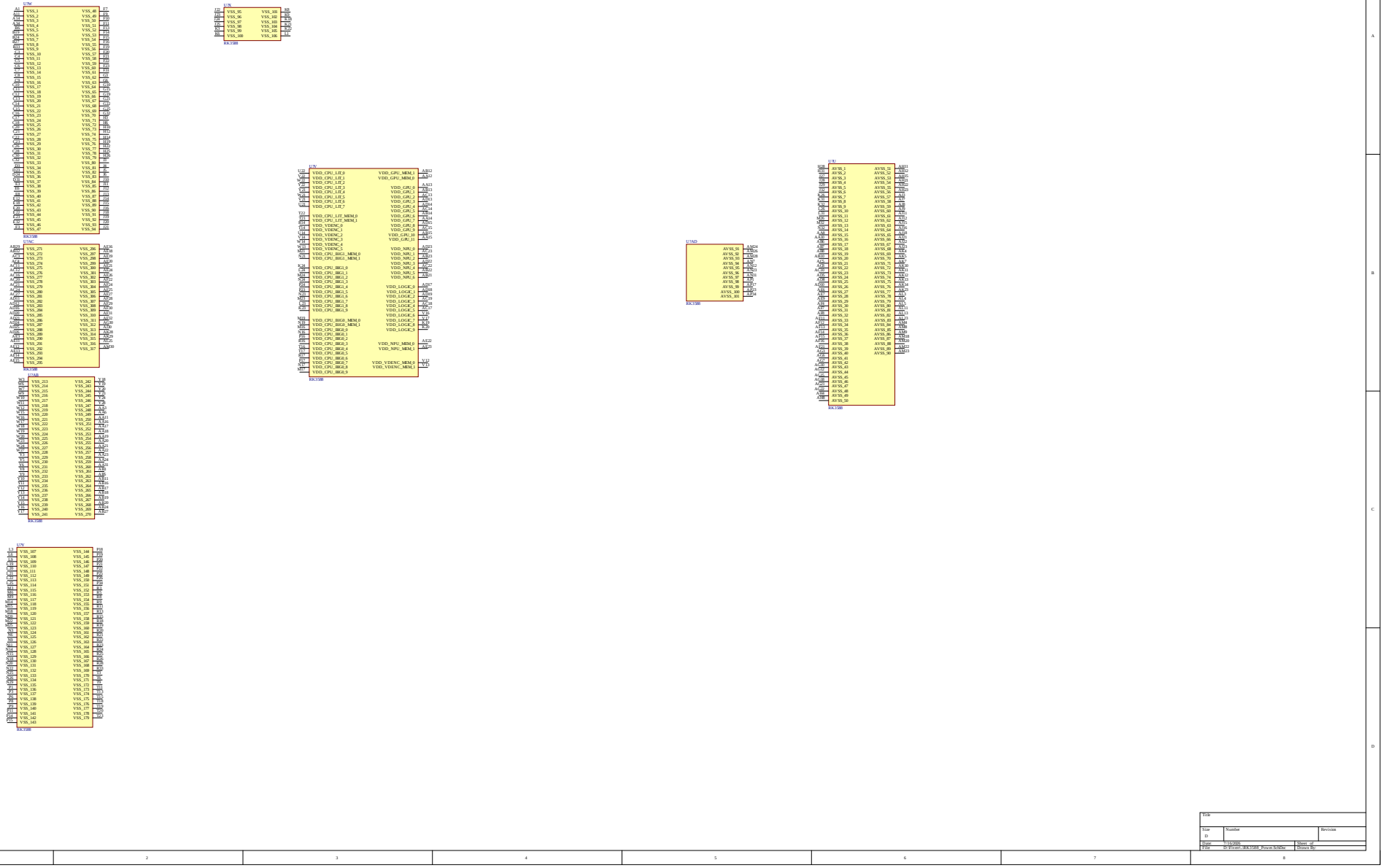
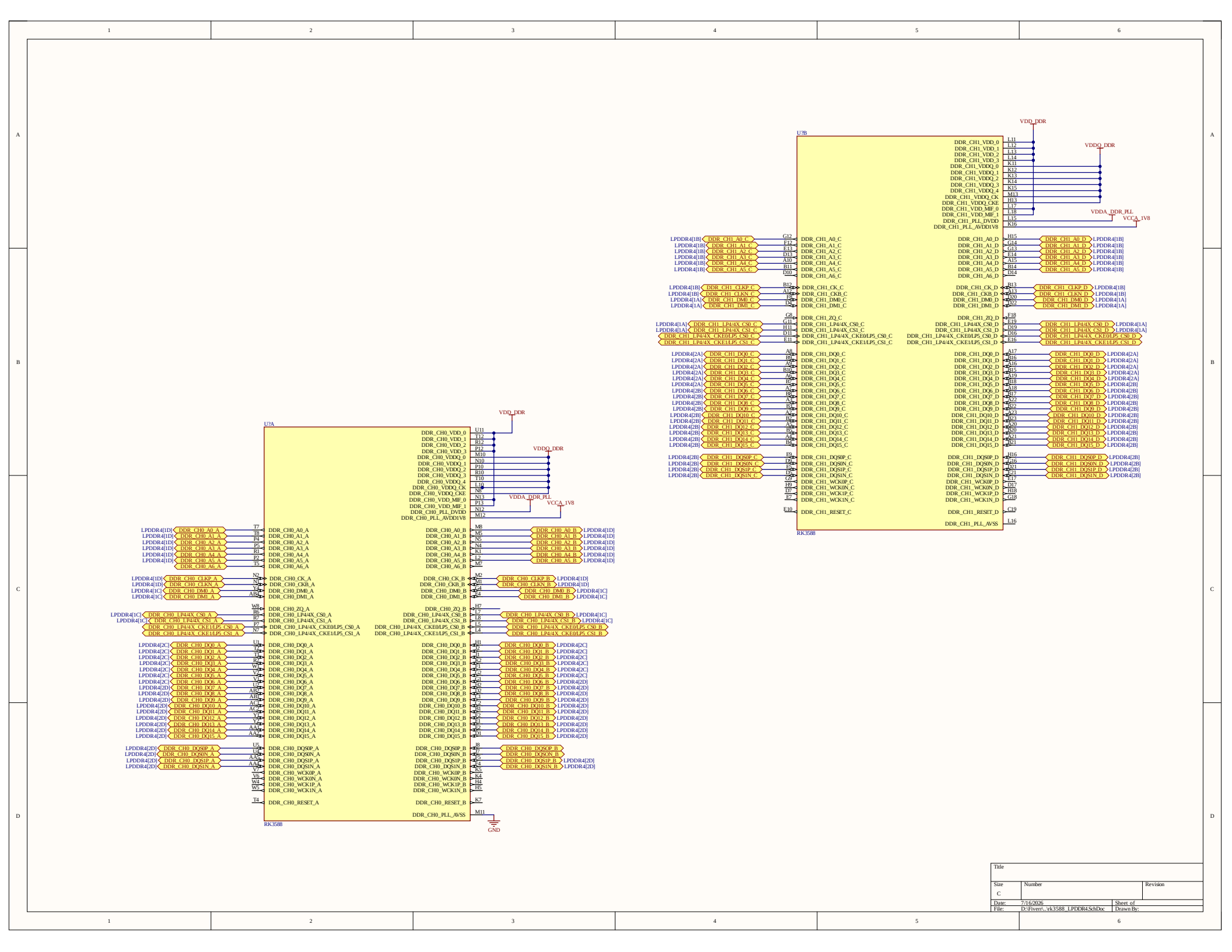
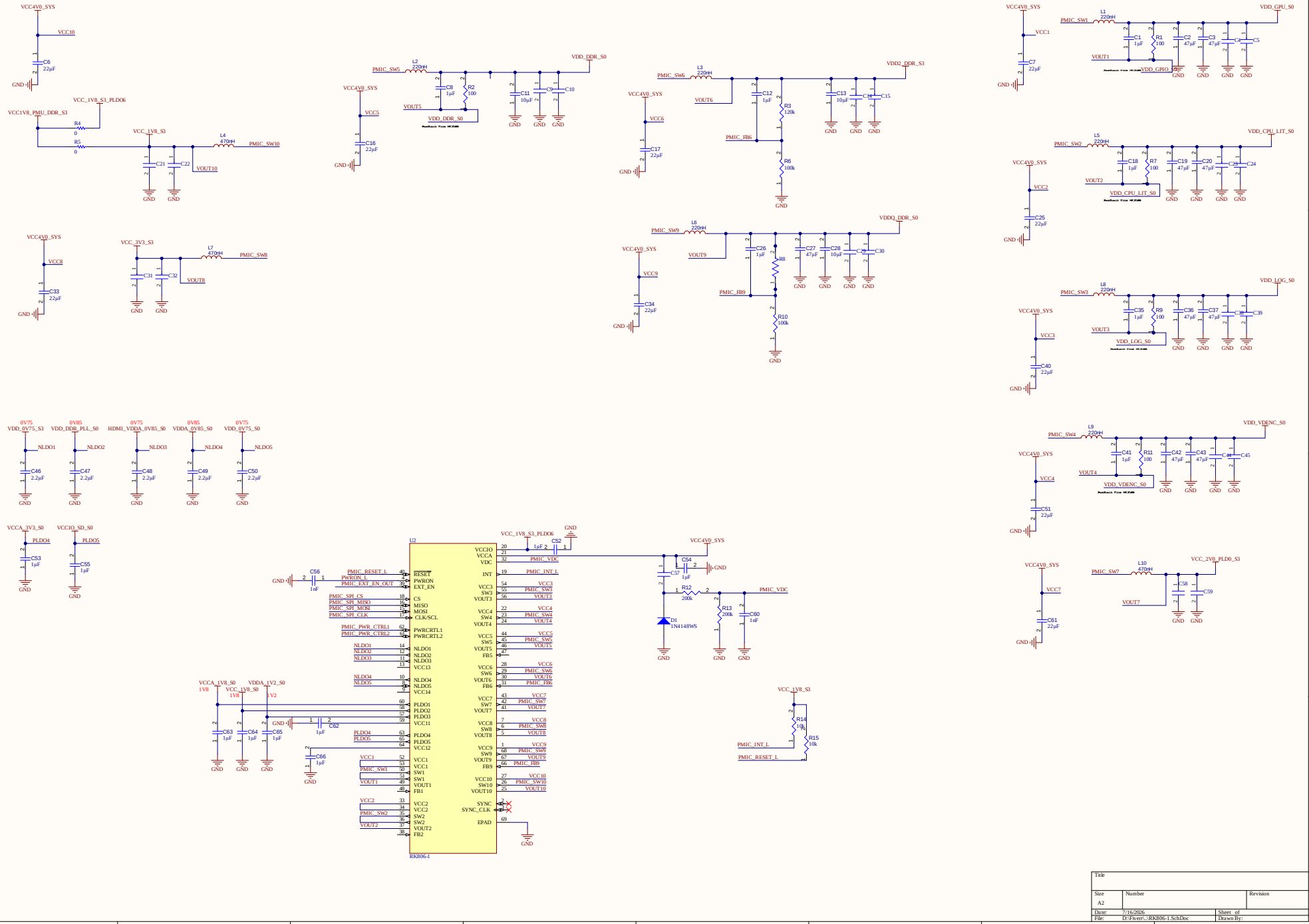
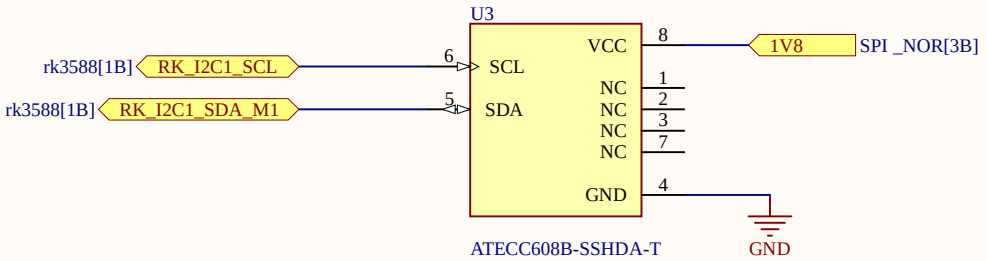


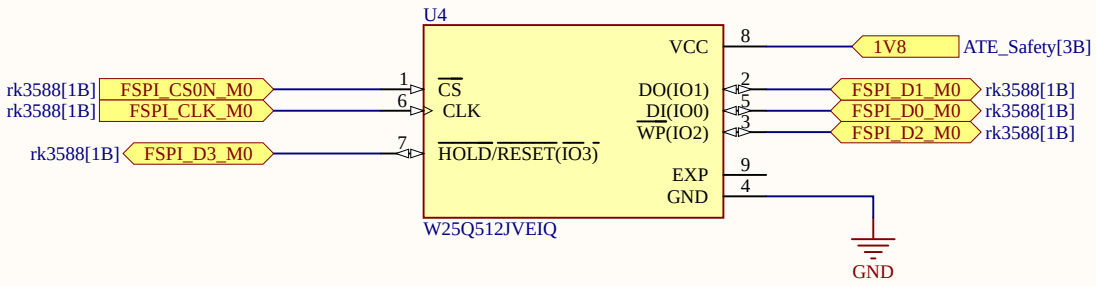

U7W
RK3588
U7AC
RK3588
U7AB
RK3588
U7Y
RK3588
U7Z
RK3588
U7V
RK3588
U7AD
RK3588
U7BJ
RK3588
Title
Size
D
Number
Revision
Date: 7/15/2025
File: D:\Project...RK3588_Power.SchDoc
Sheet of
Drawn By:


U7B
VDD_DDR
VDDQ_DDR
VDDA_DDR_PLL
VCCA_1V8
DDR_CH1_VDD_0
DDR_CH1_VDD_1
DDR_CH1_VDD_2
DDR_CH1_VDD_3
DDR_CH1_VDDQ_0
DDR_CH1_VDDQ_1
DDR_CH1_VDDQ_2
DDR_CH1_VDDQ_3
DDR_CH1_VDDQ_4
DDR_CH1_VDDQ_CK
DDR_CH1_VDDQ_CKE
DDR_CH1_VDD_MIF_0
DDR_CH1_VDD_MIF_1
DDR_CH1_PLL_DVDD
DDR_CH1_PLL_AVDD1V8
DDR_CH1_A0_C
DDR_CH1_A1_C
DDR_CH1_A2_C
DDR_CH1_A3_C
DDR_CH1_A4_C
DDR_CH1_A5_C
DDR_CH1_A6_C
DDR_CH1_CK_C
DDR_CH1_CKB_C
DDR_CH1_DM0_C
DDR_CH1_DM1_C
DDR_CH1_ZQ_C
DDR_CH1_LP4/4X_CS0_C
DDR_CH1_LP4/4X_CS1_C
DDR_CH1_LP4/4X_CKE0/LP5_CS0_C
DDR_CH1_LP4/4X_CKE1/LP5_CS1_C
DDR_CH1_DQ0_C
DDR_CH1_DQ1_C
DDR_CH1_DQ2_C
DDR_CH1_DQ3_C
DDR_CH1_DQ4_C
DDR_CH1_DQ5_C
DDR_CH1_DQ6_C
DDR_CH1_DQ7_C
DDR_CH1_DQ8_C
DDR_CH1_DQ9_C
DDR_CH1_DQ10_C
DDR_CH1_DQ11_C
DDR_CH1_DQ12_C
DDR_CH1_DQ13_C
DDR_CH1_DQ14_C
DDR_CH1_DQ15_C
DDR_CH1_DQS0P_C
DDR_CH1_DQS0N_C
DDR_CH1_DQS1P_C
DDR_CH1_DQS1N_C
DDR_CH1_WCK0P_C
DDR_CH1_WCK0N_C
DDR_CH1_WCK1P_C
DDR_CH1_WCK1N_C
DDR_CH1_RESET_C
DDR_CH1_PLL_AVSS
DDR_CH1_A0_D
DDR_CH1_A1_D
DDR_CH1_A2_D
DDR_CH1_A3_D
DDR_CH1_A4_D
DDR_CH1_A5_D
DDR_CH1_A6_D
DDR_CH1_CK_D
DDR_CH1_CKB_D
DDR_CH1_DM0_D
DDR_CH1_DM1_D
DDR_CH1_ZQ_D
DDR_CH1_LP4/4X_CS0_D
DDR_CH1_LP4/4X_CS1_D
DDR_CH1_LP4/4X_CKE0/LP5_CS0_D
DDR_CH1_LP4/4X_CKE1/LP5_CS1_D
DDR_CH1_DQ0_D
DDR_CH1_DQ1_D
DDR_CH1_DQ2_D
DDR_CH1_DQ3_D
DDR_CH1_DQ4_D
DDR_CH1_DQ5_D
DDR_CH1_DQ6_D
DDR_CH1_DQ7_D
DDR_CH1_DQ8_D
DDR_CH1_DQ9_D
DDR_CH1_DQ10_D
DDR_CH1_DQ11_D
DDR_CH1_DQ12_D
DDR_CH1_DQ13_D
DDR_CH1_DQ14_D
DDR_CH1_DQ15_D
DDR_CH1_DQS0P_D
DDR_CH1_DQS0N_D
DDR_CH1_DQS1P_D
DDR_CH1_DQS1N_D
DDR_CH1_WCK0P_D
DDR_CH1_WCK0N_D
DDR_CH1_WCK1P_D
DDR_CH1_WCK1N_D
DDR_CH1_RESET_D
DDR_CH1_PLL_AVSS
RK3588
U7A
VDD_DDR
VDDQ_DDR
VDDA_DDR_PLL
VCCA_1V8
DDR_CH0_VDD_0
DDR_CH0_VDD_1
DDR_CH0_VDD_2
DDR_CH0_VDD_3
DDR_CH0_VDDQ_0
DDR_CH0_VDDQ_1
DDR_CH0_VDDQ_2
DDR_CH0_VDDQ_3
DDR_CH0_VDDQ_4
DDR_CH0_VDDQ_CK
DDR_CH0_VDDQ_CKE
DDR_CH0_VDD_MIF_0
DDR_CH0_VDD_MIF_1
DDR_CH0_PLL_DVDD
DDR_CH0_PLL_AVDD1V8
DDR_CH0_A0_A
DDR_CH0_A1_A
DDR_CH0_A2_A
DDR_CH0_A3_A
DDR_CH0_A4_A
DDR_CH0_A5_A
DDR_CH0_A6_A
DDR_CH0_CK_A
DDR_CH0_CKB_A
DDR_CH0_DM0_A
DDR_CH0_DM1_A
DDR_CH0_ZQ_A
DDR_CH0_LP4/4X_CS0_A
DDR_CH0_LP4/4X_CS1_A
DDR_CH0_LP4/4X_CKE0/LP5_CS0_A
DDR_CH0_LP4/4X_CKE1/LP5_CS1_A
DDR_CH0_DQ0_A
DDR_CH0_DQ1_A
DDR_CH0_DQ2_A
DDR_CH0_DQ3_A
DDR_CH0_DQ4_A
DDR_CH0_DQ5_A
DDR_CH0_DQ6_A
DDR_CH0_DQ7_A
DDR_CH0_DQ8_A
DDR_CH0_DQ9_A
DDR_CH0_DQ10_A
DDR_CH0_DQ11_A
DDR_CH0_DQ12_A
DDR_CH0_DQ13_A
DDR_CH0_DQ14_A
DDR_CH0_DQ15_A
DDR_CH0_DQS0P_A
DDR_CH0_DQS0N_A
DDR_CH0_DQS1P_A
DDR_CH0_DQS1N_A
DDR_CH0_WCK0P_A
DDR_CH0_WCK0N_A
DDR_CH0_WCK1P_A
DDR_CH0_WCK1N_A
DDR_CH0_RESET_A
DDR_CH0_PLL_AVSS
DDR_CH0_A0_B
DDR_CH0_A1_B
DDR_CH0_A2_B
DDR_CH0_A3_B
DDR_CH0_A4_B
DDR_CH0_A5_B
DDR_CH0_A6_B
DDR_CH0_CK_B
DDR_CH0_CKB_B
DDR_CH0_DM0_B
DDR_CH0_DM1_B
DDR_CH0_ZQ_B
DDR_CH0_LP4/4X_CS0_B
DDR_CH0_LP4/4X_CS1_B
DDR_CH0_LP4/4X_CKE0/LP5_CS0_B
DDR_CH0_LP4/4X_CKE1/LP5_CS1_B
DDR_CH0_DQ0_B
DDR_CH0_DQ1_B
DDR_CH0_DQ2_B
DDR_CH0_DQ3_B
DDR_CH0_DQ4_B
DDR_CH0_DQ5_B
DDR_CH0_DQ6_B
DDR_CH0_DQ7_B
DDR_CH0_DQ8_B
DDR_CH0_DQ9_B
DDR_CH0_DQ10_B
DDR_CH0_DQ11_B
DDR_CH0_DQ12_B
DDR_CH0_DQ13_B
DDR_CH0_DQ14_B
DDR_CH0_DQ15_B
DDR_CH0_DQS0P_B
DDR_CH0_DQS0N_B
DDR_CH0_DQS1P_B
DDR_CH0_DQS1N_B
DDR_CH0_WCK0P_B
DDR_CH0_WCK0N_B
DDR_CH0_WCK1P_B
DDR_CH0_WCK1N_B
DDR_CH0_RESET_B
DDR_CH0_PLL_AVSS
RK3588
GND
Title
Size
C
Number
Revision
Date: 7/16/2026
Sheet of
File: D:\Fiverr\..\rk3588_LPDDR4.SchDoc
Drawn By:

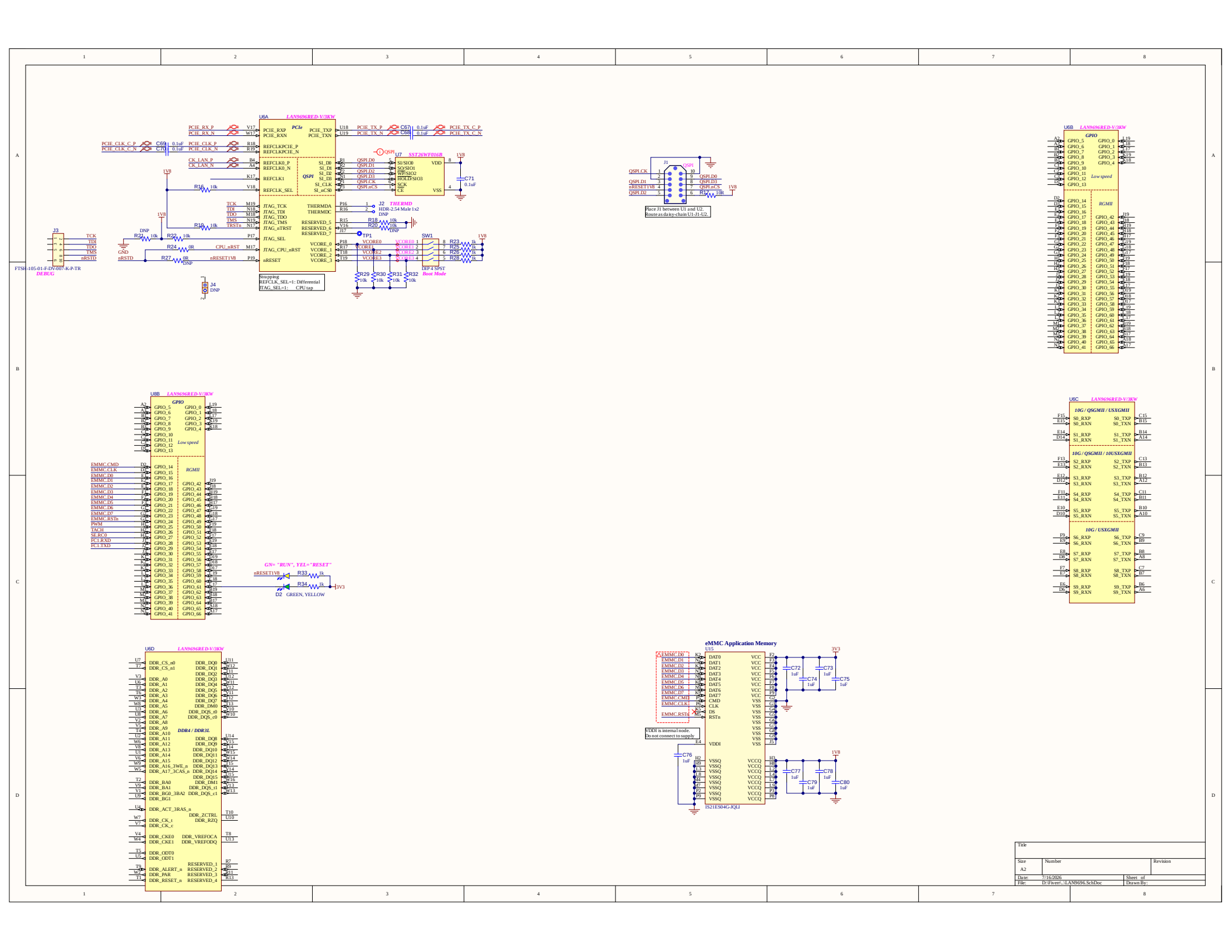

VCC4V0_SYS
VCC10
C6 22µF
GND
VCC_1V8_S3_PLDO6
VCC1V8_PMU_DDR_S3
R4 0
R5 0
VCC_1V8_S3
L4 470nH
PMIC_SW10
C21
C22
VOUT10
VDD_DDR_S0
PMIC_SW5
L2 220nH
C8 1µF
R2 100
VOUT5
C11 10µF
C9
C10
VCC5
C16 22µF
VDD2_DDR_S3
PMIC_SW6
L3 220nH
VOUT6
C12 1µF
R3 120k
C13 10µF
C14
C15
VCC6
C17 22µF
PMIC_FB6
R6 100k
VCC4V0_SYS
VCC8
C33 22µF
VCC_3V3_S3
L7 470nH
PMIC_SW8
C31
C32
VOUT8
VDDQ_DDR_S0
PMIC_SW9
L6 220nH
VOUT9
C26 1µF
R8
C27 47µF
C28 10µF
C29
C30
VCC9
C34 22µF
PMIC_FB9
R10 100k
VCC4V0_SYS
VCC1
C7 22µF
VDD_GPU_S0
PMIC_SW1
L1 220nH
C1 1µF
R1 100
C2 47µF
C3 47µF
C4
C5
VOUT1
VDD_GPIO_S0
VDD_CPU_LIT_S0
PMIC_SW2
L5 220nH
C18 1µF
R7 100
C19 47µF
C20 47µF
C23
C24
VOUT2
VCC2
C25 22µF
VDD_LOG_S0
PMIC_SW3
L8 220nH
C35 1µF
R9 100
C36 47µF
C37 47µF
C38
C39
VOUT3
VCC3
C40 22µF
VDD_VDENC_S0
PMIC_SW4
L9 220nH
C41 1µF
R11 100
C42 47µF
C43 47µF
C44
C45
VOUT4
VCC4
C51 22µF
VCC_2V0_PLD0_S3
PMIC_SW7
L10 470nH
C58
C59
VOUT7
VCC7
C61 22µF
0V75
VDD_0V75_S3
VDD_DDR_PLL_S0
HDMI_VDDA_0V85_S0
VDDA_0V85_S0
VDD_0V75_S0
NLDO1
NLDO2
NLDO3
NLDO4
NLDO5
C46 2.2µF
C47 2.2µF
C48 2.2µF
C49 2.2µF
C50 2.2µF
VCCA_3V3_S0
VCCIO_SD_S0
PLDO4
PLDO5
C53 1µF
C55 1µF
U2
RK806-1
VCCIO
VCCA
VDC
INT
VCC3
SW3
VOUT3
VCC4
SW4
VOUT4
VCC5
SW5
VOUT5
FB5
VCC6
SW6
VOUT6
FB6
VCC7
SW7
VOUT7
VCC8
SW8
VOUT8
VCC9
SW9
VOUT9
FB9
VCC10
SW10
VOUT10
SYNC
SYNC_CLK
EPAD
RESET
PWRON
EXT_EN
CS
MISO
MOSI
CLK/SCL
PWRCRTL1
PWRCRTL2
NLDO1
NLDO2
NLDO3
NLDO4
NLDO5
VCC13
VCC14
PLDO1
PLDO2
PLDO3
VCC11
PLDO4
PLDO5
VCC12
VCC1
SW1
VOUT1
FB1
VCC2
SW2
VOUT2
FB2
PMIC_RESET_L
PWRON_L
PMIC_EXT_EN_OUT
PMIC_SPI_CS
PMIC_SPI_MISO
PMIC_SPI_MOSI
PMIC_SPI_CLK
PMIC_PWR_CTRL1
PMIC_PWR_CTRL2
PMIC_INT_L
PMIC_VDC
C52 1µF
C54 1µF
C57 1µF
R12 200k
R13 200k
C60 1nF
D1 1N4148WS
C56 1nF
C62 1µF
C63 1µF
C64 1µF
C65 1µF
C66 1µF
VCCA_1V8_S0
VDDA_1V2_S0
1V8
1V2
VCC_1V8_S3
R14 10k
R15 10k
PMIC_INT_L
PMIC_RESET_L
Title
Size A2
Number
Revision
Date: 7/16/2026
Sheet of
File: D:\Fiverr\...\RK806-1.SchDoc
Drawn By:

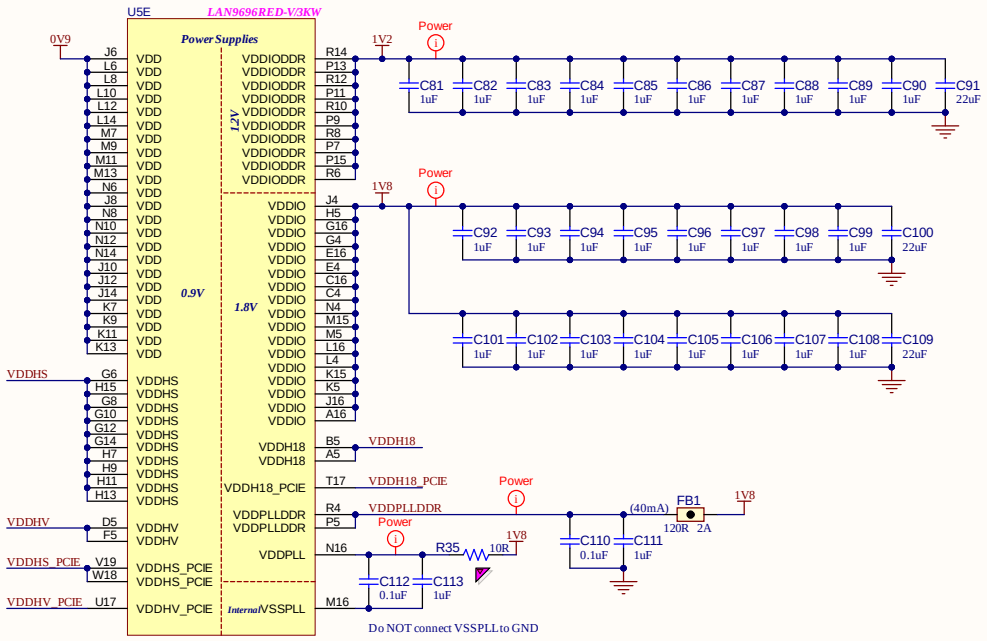



| | | |
|---|---|---|
| Title | | |
| Size: A | Number | Revision |
| Date: 7/16/2026 | | Sheet of |
| File: D:\Fiverr\..\ATE_Safety.SchDoc | | Drawn By: |

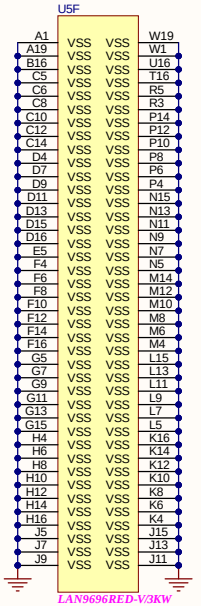


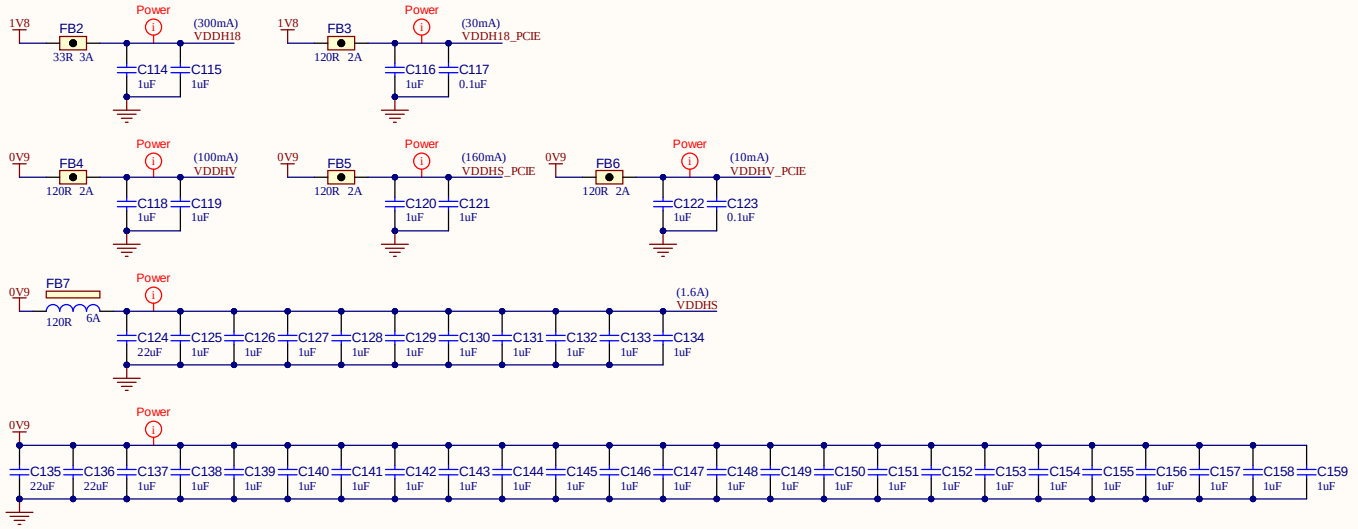

| | | | |
|---|---|---|---|
| Title | | | |
| Size A | Number | | Revision |
| Date: | 7/16/2026 | Sheet of | |
| File: | D:\Fiverr\..\SPI _NOR.SchDoc | Drawn By: | |


U6A LAN9696RED-V/3KW
PCIe
PCIE_RXP
PCIE_RXN
PCIE_TXP
PCIE_TXN
REFCLKP/CIE_P
REFCLKP/CIE_N
REFCLKO_P
REFCLKO_N
REFCLK1
REFCLK_SEL
QSPI
SI_D0
SI_DI
SI_D2
SI_D3
SI_CLK
SI_nCS0
JTAG_TCK
JTAG_TDI
JTAG_TDO
JTAG_TMS
JTAG_nTRST
JTAG_SEL
JTAG_CPU_nRST
nRESET
THERMDA
THERMDC
RESERVED_5
RESERVED_6
RESERVED_7
VCORE_0
VCORE_1
VCORE_2
VCORE_3
U7 SST26WF016B
SI/SIO0
SO/SIO1
WP/SIO2
HOLD/SIO3
SCK
CE
VDD
VSS
J1
QSPI
J2 THERMD
HDR-2.54 Male 1x2
DNP
SW1
DIP-4 SPST
Boot Mode
Strapping
REFCLK_SEL=1: Differential
JTAG_SEL=1: CPU tap
U6B LAN9696RED-V/3KW
GPIO
Low speed
RGMII
U6C LAN9696RED-V/3KW
10G / QSGMII / USXGMII
10G / QSGMII / 10USXGMII
10G / USXGMII
U8B LAN9696RED-V/3KW
GPIO
Low speed
RGMII
EMMC.CMD
EMMC.CLK
EMMC.D0
EMMC.D1
EMMC.D2
EMMC.D3
EMMC.D4
EMMC.D5
EMMC.D6
EMMC.D7
EMMC.RSTn
PWM
TACH
SER.C0
FC1.RXD
FC1.TXD
GN= "RUN", YEL="RESET"
nRESET1V8
D2 GREEN, YELLOW
U6D LAN9696RED-V/3KW
DDR4 / DDR3L
eMMC Application Memory
U15
IS21ES04G-JQLI
VDDI is internal node.
Do not connect to supply
Title
Size A2
Number
Revision
Date: 7/16/2026
Sheet of
File: D:\Fiverr\...\LAN9696.SchDoc
Drawn By:

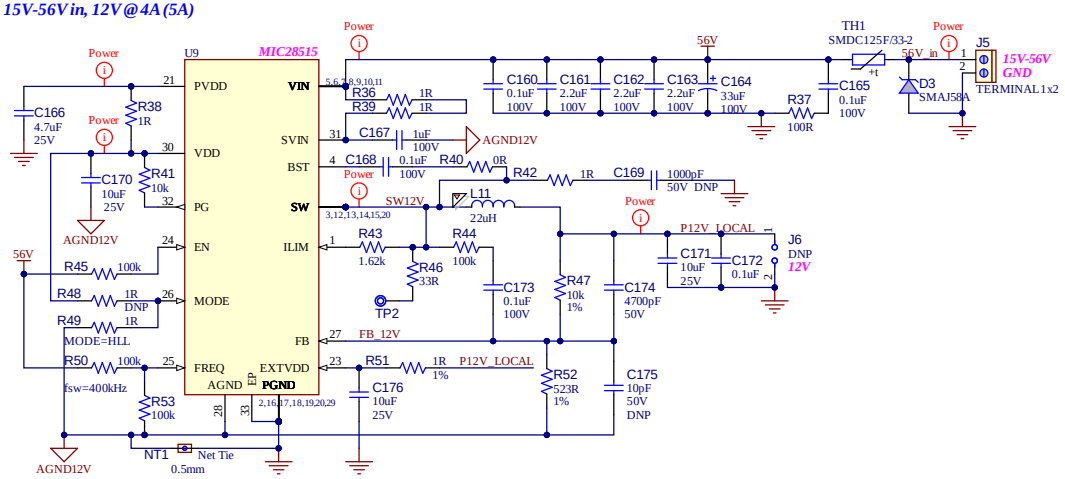



Power supplies:
0.9V 6.7A : LAN9698 VDD(5A)+VDDHS(1.6A)+VDDHV(0.1A)
1.2V 0.5A : LAN9698 VDDIODDR(0.2A)+DDR VDDQ(0.3A)
1.8V 1.7A : LAN9698 VDDH(0.3A)+VDDIO(0.5A)+ZL30732(0.5A)+Other(0.4A)

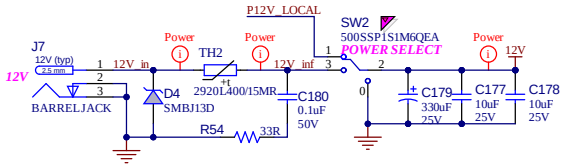


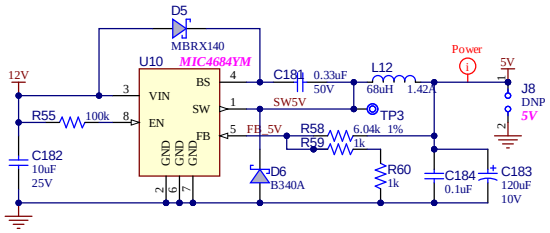
## 15V-56V in, 12V@4A (5A)



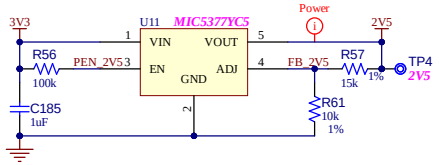

SW3 selects J19 or J23 as power source

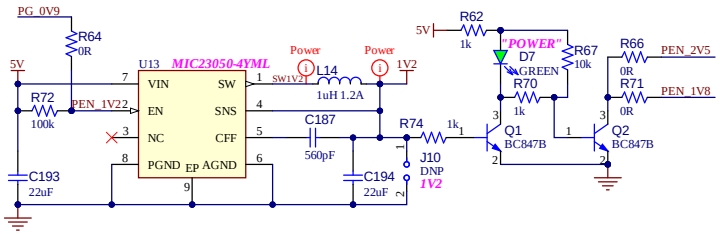


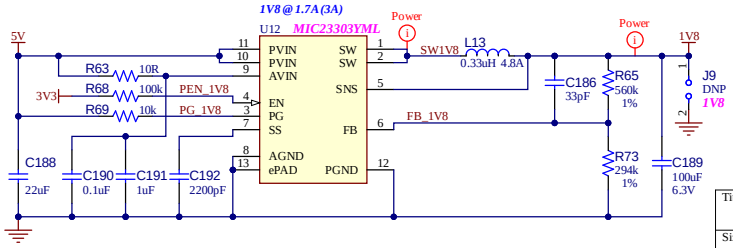

## 5V@1.2A (2A)



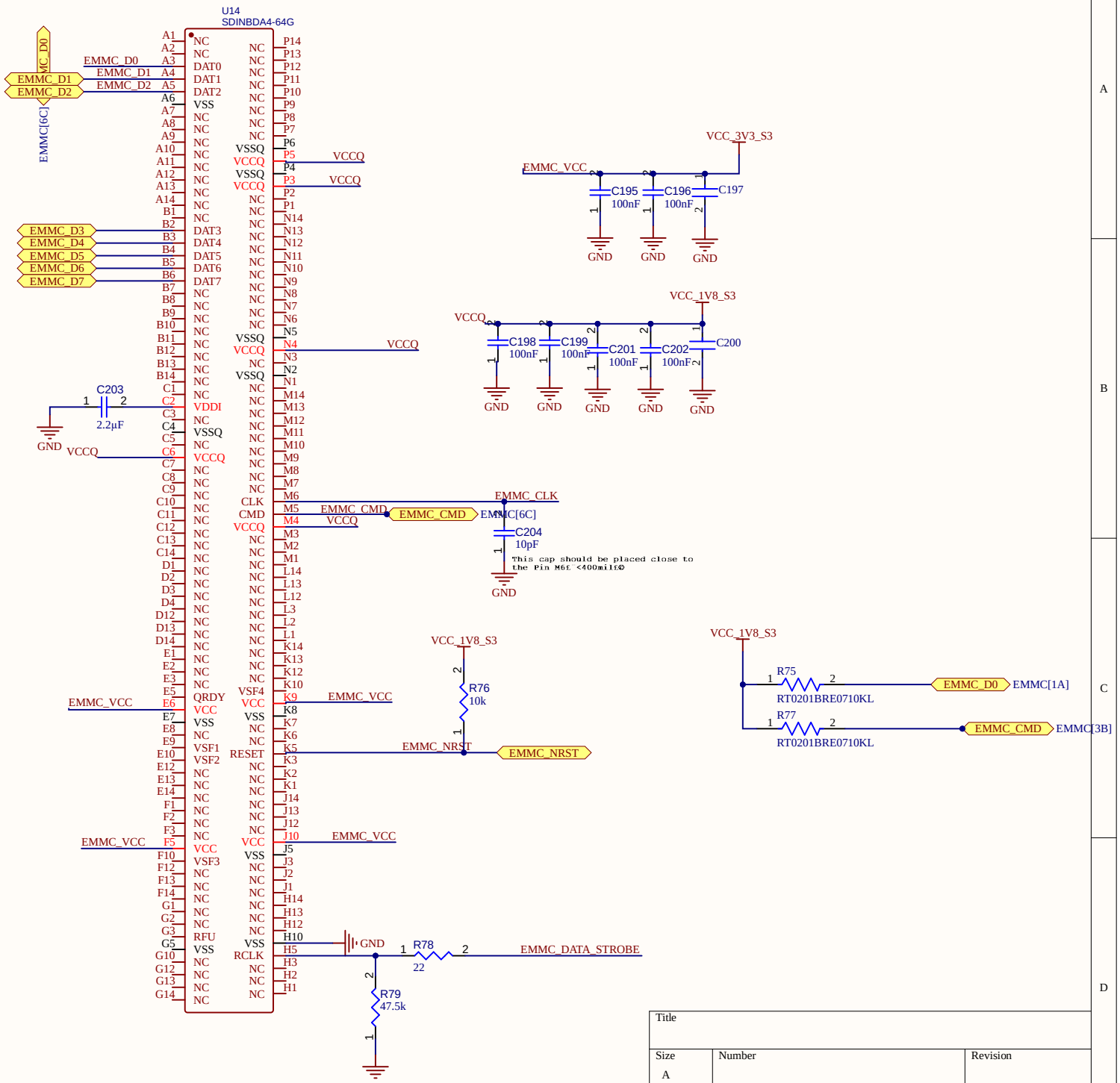

## 2V5@25mA



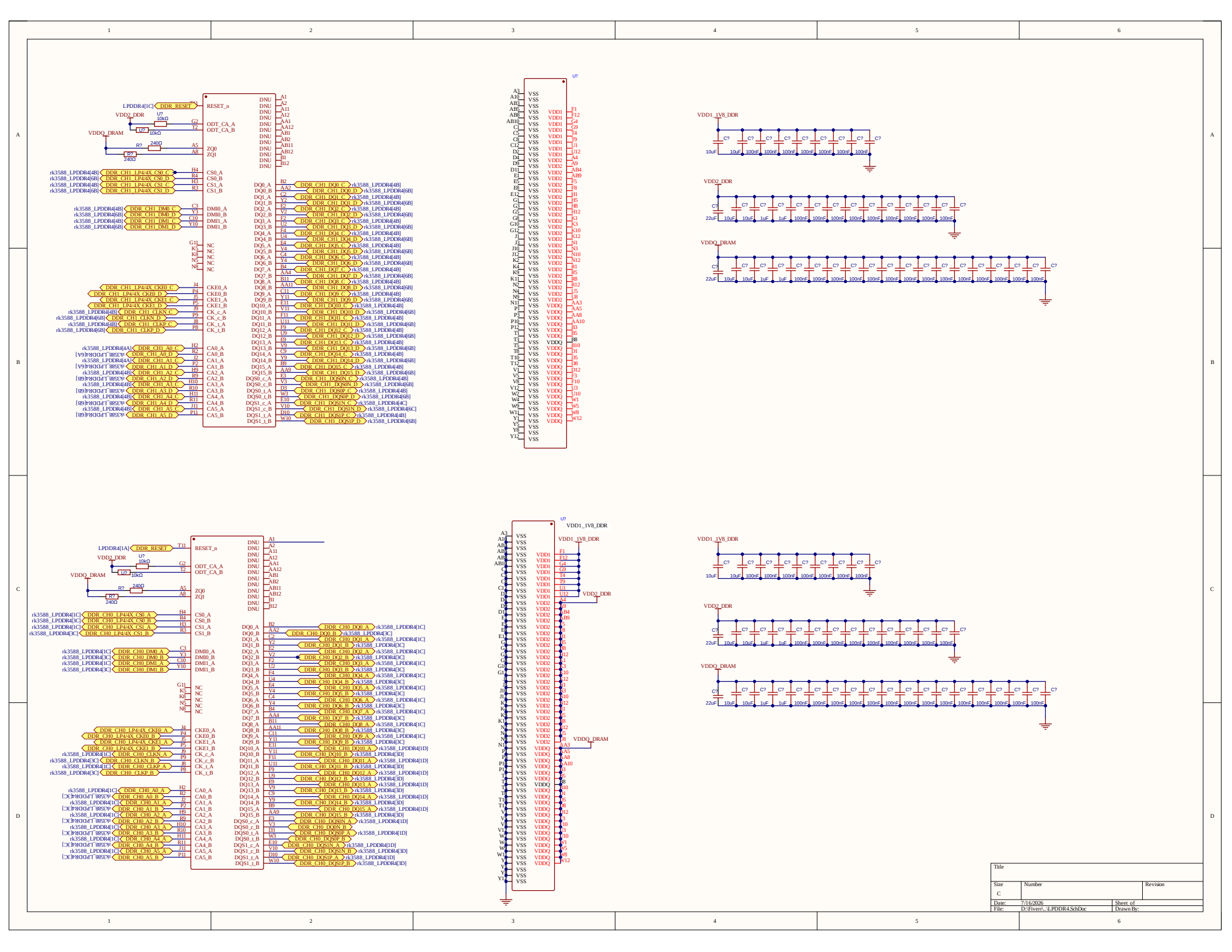




| | | |
|---|---|---|
| Title | | |
| Size A3 | Number | Revision |
| Date: 7/16/2026 | | Sheet of |
| File: D:\Fiverr\..\LAN_Supply.SchDoc | | Drawn By: |




| | | |
|---|---|---|
| Title | | |
| Size: A | Number | Revision |
| Date: 7/16/2026 | | Sheet of |
| File: D:\Fiverr\..\EMMC.SchDoc | | Drawn By: |


LPDDR4[1C]
DDR_RESET
RESET_n
VDD2_DDR
U?
10kΩ
U?
10kΩ
ODT_CA_A
ODT_CA_B
VDDQ_DRAM
R?
240Ω
ZQ0
ZQ1
R?
240Ω
rk3588_LPDDR4[4B]
DDR_CH1_LP4/4X_CS0_C
rk3588_LPDDR4[6B]
DDR_CH1_LP4/4X_CS0_D
rk3588_LPDDR4[4B]
DDR_CH1_LP4/4X_CS1_C
rk3588_LPDDR4[6B]
DDR_CH1_LP4/4X_CS1_D
CS0_A
CS0_B
CS1_A
CS1_B
rk3588_LPDDR4[4B]
DDR_CH1_DM0_C
rk3588_LPDDR4[6B]
DDR_CH1_DM0_D
rk3588_LPDDR4[4B]
DDR_CH1_DM1_C
rk3588_LPDDR4[6B]
DDR_CH1_DM1_D
DMI0_A
DMI0_B
DMI1_A
DMI1_B
NC
NC
NC
NC
NC
DDR_CH1_LP4/4X_CKE0_C
DDR_CH1_LP4/4X_CKE0_D
DDR_CH1_LP4/4X_CKE1_C
DDR_CH1_LP4/4X_CKE1_D
rk3588_LPDDR4[4B]
DDR_CH1_CLKN_C
rk3588_LPDDR4[6B]
DDR_CH1_CLKN_D
rk3588_LPDDR4[4B]
DDR_CH1_CLKP_C
rk3588_LPDDR4[6B]
DDR_CH1_CLKP_D
CKE0_A
CKE0_B
CKE1_A
CKE1_B
CK_c_A
CK_c_B
CK_t_A
CK_t_B
rk3588_LPDDR4[4A]
DDR_CH1_A0_C
rk3588_LPDDR4[6A]
DDR_CH1_A0_D
rk3588_LPDDR4[4A]
DDR_CH1_A1_C
rk3588_LPDDR4[6A]
DDR_CH1_A1_D
rk3588_LPDDR4[4B]
DDR_CH1_A2_C
rk3588_LPDDR4[6B]
DDR_CH1_A2_D
rk3588_LPDDR4[4B]
DDR_CH1_A3_C
rk3588_LPDDR4[6B]
DDR_CH1_A3_D
rk3588_LPDDR4[4B]
DDR_CH1_A4_C
rk3588_LPDDR4[6B]
DDR_CH1_A4_D
rk3588_LPDDR4[4B]
DDR_CH1_A5_C
rk3588_LPDDR4[6B]
DDR_CH1_A5_D
CA0_A
CA0_B
CA1_A
CA1_B
CA2_A
CA2_B
CA3_A
CA3_B
CA4_A
CA4_B
CA5_A
CA5_B
DDR_CH1_DQ0_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ0_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ1_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ1_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ2_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ2_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ3_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ3_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ4_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ4_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ5_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ5_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ6_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ6_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ7_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ7_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ8_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ8_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ9_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ9_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ10_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ10_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ11_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ11_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ12_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ12_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ13_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ13_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ14_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ14_D
rk3588_LPDDR4[6B]
DDR_CH1_DQ15_C
rk3588_LPDDR4[4B]
DDR_CH1_DQ15_D
rk3588_LPDDR4[6B]
DDR_CH1_DQS0N_C
rk3588_LPDDR4[4B]
DDR_CH1_DQS0N_D
rk3588_LPDDR4[6B]
DDR_CH1_DQS0P_C
rk3588_LPDDR4[4B]
DDR_CH1_DQS0P_D
rk3588_LPDDR4[6B]
DDR_CH1_DQS1N_C
rk3588_LPDDR4[4C]
DDR_CH1_DQS1N_D
rk3588_LPDDR4[6C]
DDR_CH1_DQS1P_C
rk3588_LPDDR4[4B]
DDR_CH1_DQS1P_D
rk3588_LPDDR4[6B]
DQ0_A
DQ0_B
DQ1_A
DQ1_B
DQ2_A
DQ2_B
DQ3_A
DQ3_B
DQ4_A
DQ4_B
DQ5_A
DQ5_B
DQ6_A
DQ6_B
DQ7_A
DQ7_B
DQ8_A
DQ8_B
DQ9_A
DQ9_B
DQ10_A
DQ10_B
DQ11_A
DQ11_B
DQ12_A
DQ12_B
DQ13_A
DQ13_B
DQ14_A
DQ14_B
DQ15_A
DQ15_B
DQS0_c_A
DQS0_c_B
DQS0_t_A
DQS0_t_B
DQS1_c_A
DQS1_c_B
DQS1_t_A
DQS1_t_B
U?
VDD1_1V8_DDR
VDD2_DDR
VDDQ_DRAM
VDD1_1V8_DDR
VDD2_DDR
VDDQ_DRAM
C?
10uF
22uF
100nF
1uF
Title
Size
C
Number
Revision
Date:
7/16/2026
Sheet of
File:
D:\Fiverr\..\LPDDR4.SchDoc
Drawn By: